

REPORT

Part 1

```
(Setting up nodejs (16.20.2-Inodesource1) ...  
Processing triggers for man-db (2.10.2-1) ...  
user@user-HP-EliteBook-830-G6:~$ node -v  
v16.20.2  
user@user-HP-EliteBook-830-G6:~$ npm -v  
8.19.4  
user@user-HP-EliteBook-830-G6:~$
```

```
user@user-HP-EliteBook-830-G6:~/Desktop$ git clone https://github.com/zaheersani/SCD-SP25-NodeApp.git  
Cloning into 'SCD-SP25-NodeApp'...  
remote: Enumerating objects: 9, done.  
remote: Counting objects: 100% (9/9), done.  
remote: Compressing objects: 100% (8/8), done.  
remote: Total 9 (delta 0), reused 9 (delta 0), pack-reused 0 (from 0)  
Receiving objects: 100% (9/9), done.  
user@user-HP-EliteBook-830-G6:~/Desktop$ ls  
SCD_ASSIGNMENT  SCD-SP25-NodeApp  
user@user-HP-EliteBook-830-G6:~/Desktop$ cd SCD-SP25-NodeApp  
user@user-HP-EliteBook-830-G6:~/Desktop/SCD-SP25-NodeApp$ ls  
app.js  package.json  
user@user-HP-EliteBook-830-G6:~/Desktop/SCD-SP25-NodeApp$ npm install  
npm WARN EBADENGINE Unsupported engine {  
npm WARN EBADENGINE   package: 'express@5.1.0',  
npm WARN EBADENGINE   required: { node: '>= 18' },  
npm WARN EBADENGINE   current: { node: 'v16.20.2', npm: '8.19.4' }  
npm WARN EBADENGINE }  
npm WARN EBADENGINE Unsupported engine {  
npm WARN EBADENGINE   package: 'body-parser@2.2.0',  
npm WARN EBADENGINE   required: { node: '>=18' },  
npm WARN EBADENGINE   current: { node: 'v16.20.2', npm: '8.19.4' }  
npm WARN EBADENGINE }
```

```
user@user-HP-EliteBook-830-G6:~/Desktop/SCD-SP25-NodeApp$ node app.js &  
[1] 48371  
user@user-HP-EliteBook-830-G6:~/Desktop/SCD-SP25-NodeApp$ Server is running on http://localhost:3000
```

```
localhost:3000/todo/1  
Pretty-print ☒  
{  
  "error": "Internal Server Error"  
}
```

Deployment Error Summary

During the deployment of the backend Node.js application from the GitHub repository <https://github.com/zaheersani/SCD-SP25-NodeApp>, an error was encountered while accessing the endpoint `/todo/1`. The server responded with a JSON message stating "Internal Server Error", indicating that the application failed to handle the request correctly. Upon investigation, it was found that the root cause of this issue was an environment mismatch between the development and production setups, specifically related to the Node.js version.

Node.js Version Incompatibility

The application appears to have been developed using a newer version of Node.js (most likely Node 18 or above), while the production server has Node.js 16 installed. This mismatch can result in runtime failures because newer JavaScript syntax and features—such as ES modules (`import/export`), optional chaining (`?.`), and top-level `await`—are not supported in Node.js 16. Additionally, the `package-lock.json` file generated during development may cause dependency resolution issues when installed under a different Node.js version, leading to application crashes or incorrect behavior.

Dependency and Runtime Issues

The application's dependencies may include packages that require Node.js 18 or higher to function properly. For instance, updated versions of libraries such as `mongoose`, `express`, and `dotenv` could rely on newer Node.js features or native bindings that are not compatible with Node 16. When these packages are run in an unsupported environment, errors like `TypeError: x is not a function` or `SyntaxError: Unexpected token` may occur, which can cause the application to fail during runtime and result in "Internal Server Error" responses to API requests.

Invalid Request Input: MongoDB ObjectID

Another important issue was observed in the test request made to the endpoint `/todo/1`. The value `"1"` is not a valid MongoDB ObjectID. MongoDB requires a 24-character hexadecimal string as the identifier for documents. When the application receives an invalid ID like `"1"`, Mongoose attempts to cast it and throws an internal error, which is not handled gracefully in the code. This also contributes to the 500 Internal Server Error seen in the browser and highlights a need for proper validation and error handling in the application.

Part 2

```
sion denied
user@user-HP-EliteBook-830-G6:~/Desktop/SCD-SP25-NodeApp$ sudo docker build -t scd-nodeapp:v1 .
[sudo] password for user:
[+] Building 1345.4s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                  0.0s
=> => transferring dockerfile: 222B                                  0.0s
=> [internal] load metadata for docker.io/library/node:18           7.7s
=> [internal] load .dockerignore                                     0.0s
=> => transferring context: 67B                                       0.0s
=> [1/5] FROM docker.io/library/node:18@sha256:867be01f97d45cb7d89a8e 1329.4s
=> => resolve docker.io/library/node:18@sha256:867be01f97d45cb7d89a8ef0b 0.0s
=> => sha256:a0ee20d21ff7e3199f16644fd076ef1435b42689ca0 2.49kB / 2.49kB 0.0s
=> => sha256:e56a6e3bc5724c51df0a1ad86c6e38aee233bb34345 6.39kB / 6.39kB 0.0s
=> => sha256:63964a8518f54dc31f8df89d7f06714c7a793aa 24.01MB / 24.01MB 198.1s
=> => sha256:ca513cad20bb13ead2c745499459eed59a6db34 64.39MB / 64.39MB 674.1s
```

```
app.js Dockerfile node_modules package.json package-lock.json
user@user-HP-EliteBook-830-G6:~/Desktop/SCD-SP25-NodeApp$ docker run -d -p 3000:3000 --name scd-container scd-nodeapp:v1
docker: permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Head "http://%2Fvar%2Frun%2Fdocker.sock/_ping": dial unix /var/run/docker.sock: connect: permission denied

Run 'docker run --help' for more information
user@user-HP-EliteBook-830-G6:~/Desktop/SCD-SP25-NodeApp$ sudo docker run -d -p 3000:3000 --name scd-container scd-nodeapp:v1
[sudo] password for user:
0a68f66caaa8c4643d9929b92ad5ac6b1376554b046ebf0ebd6f061fe501c406
user@user-HP-EliteBook-830-G6:~/Desktop/SCD-SP25-NodeApp$
```

```
localhost:3000/todo/0a68f66caaa8c4643d9929b92ad5ac6b1376554b046ebf0ebd6f061fe501c406
```

Pretty-print ☐

```
{}
```

```
user@user-HP-EliteBook-830-G6:~/Desktop/SCD-SP25-NodeApp$ curl http://localhost:3000/todo/662ff74b53dd2d6d24a89dbf
{}
user@user-HP-EliteBook-830-G6:~/Desktop/SCD-SP25-NodeApp$ docker push yourusername/scd-nodeapp:v1
```

```

user@user-HP-EliteBook-830-G6: ~/Desktop/SCD-SP25-NodeApp$ newgrp docker
user@user-HP-EliteBook-830-G6: ~/Desktop/SCD-SP25-NodeApp$ docker login -u sabeehdayyan

Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit https://app.docker.com/settings

Password:
Login Succeeded
user@user-HP-EliteBook-830-G6: ~/Desktop/SCD-SP25-NodeApp$ docker push sabeehdayyan/scd-nodeapp:v1
The push refers to repository [docker.io/sabeehdayyan/scd-nodeapp]
ef41a4625edb: Pushed
995f75f89853: Pushed
d98adb743e30: Pushed
f2ef8957f471: Pushed
4ef4dcd20f76: Mounted from library/node
438e733c9a0c: Mounted from library/node
f7bcf5aff5f1: Mounted from library/node
2f3ebdcf6a27: Mounted from library/node
bf9c09fb6f3a: Mounted from library/node
fcbb8c0ae5d6: Mounted from library/node
8ce3e08e661a: Mounted from library/node
247ffffb7158d: Mounted from library/node
v1: digest: sha256:68348b6e54eaa64637f3544d179615205bc1ffad5c8856bbc2b246314a4af49d size: 2838
user@user-HP-EliteBook-830-G6: ~/Desktop/SCD-SP25-NodeApp$

```

```

user@user-HP-EliteBook-830-G6: ~/Desktop/SCD-SP25-NodeApp$ docker run -d -p 3001:3000 --name nodeapp-container sabeehdayyan/scd-nodeapp:v1
ef0ff98ed0171f6dd1358578d2f2468bca5ad21f805a2aa19a26820d3d074ca1
user@user-HP-EliteBook-830-G6: ~/Desktop/SCD-SP25-NodeApp$ curl http://localhost:6000/todo/ef0ff98ed0171f6dd1358578d2f2468bca5ad21f805a2aa19a26820d3d074ca1
curl: (7) Failed to connect to localhost port 6000 after 0 ms: Connection refused
user@user-HP-EliteBook-830-G6: ~/Desktop/SCD-SP25-NodeApp$ curl http://localhost:3001/todo/ef0ff98ed0171f6dd1358578d2f2468bca5ad21f805a2aa19a26820d3d074ca1
user@user-HP-EliteBook-830-G6: ~/Desktop/SCD-SP25-NodeApp$

```

```

▼ TERMINAL
● user@user-HP-EliteBook-830-G6: ~/Desktop/assignment/backend$ git branch
  main
* user
● user@user-HP-EliteBook-830-G6: ~/Desktop/assignment/backend$ git checkout main
M      backend/package-lock.json
M      backend/package.json
M      backend/server.js
M      frontend/src/Home.js
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
● user@user-HP-EliteBook-830-G6: ~/Desktop/assignment/backend$ git pull origin main
From https://github.com/s-a-b-e-e-h/assignment
* branch      main      -> FETCH_HEAD
Already up to date.
● user@user-HP-EliteBook-830-G6: ~/Desktop/assignment/backend$ git merge user
Already up to date.
● user@user-HP-EliteBook-830-G6: ~/Desktop/assignment/backend$ git push origin main
Everything up-to-date
● user@user-HP-EliteBook-830-G6: ~/Desktop/assignment/backend$ git branch
* main
  user
○ user@user-HP-EliteBook-830-G6: ~/Desktop/assignment/backend$

```